DAC VINH NGUYEN

AI Engineer · MSc Computational Finance
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Eligible for a UK Graduate visa (2 years); no sponsorship needed to start.

PROFILE

AI engineer who built and shipped two desktop apps in Python: GoodLuck, with an LLM assistant (Claude API, Groq) and 5,500+ tests, and Cat Studio. MSc Computational Finance, CFA Level I.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, SciPy, PyTorch), C++17 (std::thread, ctypes bindings, ThreadSanitizer), SQL (DuckDB), Git
Machine learning: gradient boosting, logistic regression, LSTM, transformer fine-tuning (DistilBERT), cross-validation, walk-forward validation, fairness checks
Quantitative finance: Value-at-Risk and Expected Shortfall, GARCH and GJR-GARCH, Kupiec and Christoffersen backtests, Black-Scholes, binomial trees, Monte Carlo, portfolio optimisation
Data: incremental loading into DuckDB, audit trails, SQL data-quality checks, dashboards in Plotly, time series, statistical inference

EDUCATION

MSc Computational Finance, Royal Holloway, University of London 2025 – 2026
Investment and Portfolio Management 86 · Data Analysis 83 · Deep Learning 83 · Principles of Computation and Programming 75

BA (Hons) Advanced Finance, National Economics University, Vietnam awarded Sep 2025
GPA 3.59 / 4.0

CFA Level I, top 10% of global candidates, October 2024 · IBM Data Science Professional Certificate · IBM Machine Learning

SELECTED PROJECTS

Each title links to the code and a one-page write-up.

GoodLuck

- Built a Windows and Mac desktop app that runs a job search end to end: reads job boards, filters by custom rules, scores jobs against a CV sentence by sentence, tracks Gmail, and fills forms in Chrome.
- Built an LLM chat assistant (Groq or Claude via Anthropic API) with guardrails so every number comes from app counts; used Python, SQLite, pywebview, Nuitka, a Telegram bot, and 5,500+ tests.

Cat Studio

- Built a Windows desktop app where users connect blocks to design a trading strategy, generate the QuantConnect LEAN algorithm in Python, and backtest it on historical data, with 26 guided levels.
- Developed a parameter optimiser with 27 metrics and 164 quantities mapped to LEAN; used Python, a web UI in a native window, .NET 10 and pythonnet.

Option pricing engine in C++

- Priced European, American and Asian options three ways in C++17: Black-Scholes with Greeks, a 2,000-step binomial tree, and a multithreaded Monte Carlo with antithetic variates, called from Python through ctypes.
- Reached 161 million paths a second on eight threads, 2.3 times vectorised NumPy, with the threading verified under ThreadSanitizer.

Value-at-Risk and Expected Shortfall, backtested over twenty years

- Compared five models on twenty years of FTSE 100 and S&P 500 returns, judged from 2007 over 4,979 trading days, every forecast using only the data available that evening.
- Found GARCH-t breaches least, 1.5% against a 1% target, and reported that Kupiec rejects all five rather than presenting the flattering one.

Day-ahead forecasting of Britain's electricity demand

- Forecast half-hourly national demand from 309,550 half-hours of history with weather and holiday features: gradient boosting misses by 3.2% over 971 test days, against 8.2% for the seasonal naive.
- A typical miss of 794 MW on a system averaging 26 GW, holding at 4.0% once real weather-forecast error is fed in, and ahead of an LSTM in PyTorch that fell behind as embedded solar grew.

Financial news sentiment with a fine-tuned transformer

- Fine-tuned DistilBERT on 4,836 annotated sentences, with near-duplicate sentences kept on one side of the split: 86% accurate against 76% for TF-IDF, and 98% where every annotator agreed.
- Macro-F1 0.85 over three seeds, widest on the rare negative class at 0.86 against 0.64, with 93% of the errors on sentences the annotators themselves split over.

Market-data warehouse

- Loaded 221,287 daily prices for 75 London and New York securities since 2015 into DuckDB incrementally, with an audit trail, 13 SQL quality checks and a one-page dashboard.
- Caught Yahoo applying the dividends of pence-quoted shares at a hundredth of their size, and rewrote the bad-print check to measure each security against its own market after it flagged real COVID moves.

EXPERIENCE

Research Consultant, WorldQuant (BRAIN platform) Jan 2025 – Sep 2025

- Generated systematic alpha signals in Python on WorldQuant's BRAIN platform.
- Scored every signal on Sharpe ratio, maximum drawdown, expected payoff and monthly risk and return rather than on profit, which on its own proves nothing: an edge can be faked.